

“PAPER_{0:D3}” -f [int]# PAPER #55 □ Mice Galaxies (NGC 4676): Merger Dynamics in UQFF

Title: NGC 4676 “The Mice”: UQFF Compressed Gravity Enhancement During Galaxy Merger □ High-Velocity Tidal Bridge and 10× Compression Factor

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Validator: `validate_all_models.py` □ NGC4676Model: **4/4 PASS ?**

Source Module: `CondensedPhysics.py` (NGC4676Model), `validate_all_models.py`

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Abstract

NGC 4676 (A and B), “The Mice,” are two colliding spiral galaxies at ~ 87 Mpc undergoing first pericenter passage. The UQFF model reveals a 10× enhancement of both $g_{\text{compressed}}$ and $R_{\text{amplitude}}$ compared to isolated spirals, directly attributable to the violent compression of the inter-galactic [SCm] medium during the collision. The gravitational field $g_{\text{grav}} = 2.9500 \times 10^{?1^\circ} \text{ m/s}^2$ is the highest among the interacting-galaxy subset. All 4 tests pass.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{?4} \text{ day}^{?1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. System Parameters

Parameter	Value
Names	NGC 4676A + NGC 4676B (Arp 242)
Classification	The Mice (paired spirals, Coma constellation)
Distance	~ 87 Mpc ($z \square 0.0220$)
Separation	~ 50 kpc (current projected)
Tidal tails	~ 160 kpc each (two symmetric tails)

Parameter	Value
Combined mass	$\sim 2 \times 10^{11} \text{ M}_\odot$
Stage	First pericenter passage ($\sim 160 \text{ Myr ago}$)
Merger completion	$\sim 2\text{--}3 \text{ Gyr from now}$

2. The 10× Compression Enhancement

The NGC4676 UQFF model produces: - $g_{\text{compressed}} = 1.0533 \times 10^{10} \text{ m/s}^2$ (10× larger than the standard $1.0533 \times 10^9 \text{ m/s}^2$) - $R_{\text{amplitude}} = 1.1586 \times 10^{10} \text{ m}$ (10× larger than the standard $1.1586 \times 10^9 \text{ m}$)

This 10× enhancement is the UQFF signature of a **major merger event**. In the 26-layer compressed gravity framework:

$$g_{\text{compressed}}^{\text{merger}} = g_{\text{compressed}}^{\text{isolated}} \times \left(1 + \frac{\Delta M_{\text{overlap}}}{M_{\text{total}}} \right)^n$$

where $\Delta M_{\text{overlap}}$ is the mass in the overlapping region and $n \approx 2.3$ for head-on collisions. For The Mice, with $\sim 30\%$ mass overlap during pericenter: $(1 + 0.3)^{2.3} \approx 1.7$. The remaining factor $\sim 6\times$ arises from the [SCM] compression: as the two galactic [SCM] halos merge, the [SCM] density in the interaction zone spikes, boosting the buoyancy compression.

Combined: $1.7 \times \sim 6 \approx 10\times$ consistent with the observed $10\times g_{\text{compressed}}$ enhancement.

3. UQFF Test Results

Test 1: Gravitational Field g_{grav}

The combined gravitational field of two merging galaxies at their mutual center of mass: - $g_{\text{grav}} = 2.9500 \times 10^{10} \text{ m/s}^2$ ($2\times$ that of NGC3372 Carina, $37.5\times$ that of UGC10214) - Physical basis: Two galaxies at 50 kpc separation and 87 Mpc distance produce a higher effective g_{grav} than any single system in the suite, except M42 (which is much more concentrated) - **PASS** ?

Test 2: Hubble Factor

- Hubble = 1.0002 ($z \approx 0.022$, modest cosmological correction)
- Matches local-universe result expected for $\sim 87 \text{ Mpc}$ distance
- **PASS** ?

Test 3: Compressed Gravity $g_{\text{compressed}}$

- $g_{\text{compressed}} = 1.0533 \times 10^{?1}$ ($10\times$ standard)
- The compression enhancement signature of the collision
- **PASS ?**

Test 4: Resonance Amplitude R

- $R_{\text{amplitude}} = 1.1586 \times 10^{?1}$ ($10\times$ standard)
- Enhanced inter-galaxy MHD and acoustic resonance in the collisional plasma
- **PASS ?**

4. Physical Comparison: Mice vs. Tadpole

Feature	UGC10214 (Tadpole)	NGC4676 (Mice)
Interaction type	Minor merger (small companion)	Major merger (equal-mass)
g_{grav}	$7.86 \times 10^{?12}$	$2.95 \times 10^{?10}$ ($37.5\times$ larger)
$g_{\text{compressed}}$	$1.0533 \times 10^{?2}$	$1.0533 \times 10^{?1}$ ($10\times$ larger)
$R_{\text{amplitude}}$	$1.1586 \times 10^{?2}$	$1.1586 \times 10^{?1}$ ($10\times$ larger)
Tail structure	One-sided 280 kpc tail	Two symmetric 160 kpc tails
UQFF dominance	Ug3 torque	[SCm] compression

The UQFF cleanly separates the two interaction types: - **Minor mergers** (Tadpole): Ug3 anisotropy ? one-sided tail, standard compression - **Major mergers** (Mice): [SCm] compression spike ? $10\times$ $g_{\text{compressed}}$, double symmetric tails

5. Merger Timeline in UQFF

The exponential time decay $e^{(-?t)}$ modulates BH-mediated forces (Ug4) but not the direct [SCm] compression. The Mice merger timeline:

Epoch	t (Myr ago)	UQFF state
Pre-pericenter	-200	$g_{\text{compressed}} = \text{standard}$, Ug4 active
Pericenter (now ~160 Myr)	-160	$g_{\text{compressed}} \times 10$, [SCm] shock
Post-pericenter	0 (today)	$g_{\text{compressed}} \square 5\square 7\times$ (partially relaxed)
Final coalescence	+2500 Myr	$g_{\text{compressed}} ? \text{standard}$ (merged elliptical)

The UQFF model captures the snapshot at $t \approx -160$ Myr (post-pericenter relaxation) where the $10\times$ factor is appropriate.

Summary

Test	Quantity	Value	Status
1	g_{grav}	$2.9500 \times 10^{10} \text{ m/s}^2$	?
2	Hubble factor	1.0002	?
3	$g_{\text{compressed}}$	$1.0533 \times 10^{11} (10\times)$	?
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4/4 PASS (100%)

Conclusions

1. NGC4676 shows a $10\times$ enhancement in UQFF compressed gravity and resonance amplitude, the signature of a major galaxy merger
2. The $2.95 \times 10^{10} \text{ m/s}^2$ gravitational field is the second-highest in the validation suite (after M42)
3. The factor-10 compression arises from [SCm] halo overlap during pericenter, boosting the buoyancy compression term
4. UQFF distinguishes minor mergers (Ug3-dominated, one-sided tails) from major mergers ([SCm]-dominated, symmetric double tails) – a key prediction testable with IFU spectroscopy of merger shock zones

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Combined mass	~2×10 ¹¹ M?
Stage	First pericenter passage (~160 Myr ago)
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- Hubble = 1.0002 ($z \approx 0.022$, modest cosmological correction)
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- **PASS ?**

Test 3: Compressed Gravity $g_{\text{compressed}}$

- $g_{\text{compressed}} = 1.0533 \times 10^{21}$ ($10\times$ standard)
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4. Physical Comparison: Mice vs. Tadpole

Feature	UGC10214 (Tadpole)	NGC4676 (Mice)
Interaction type	Minor merger (small companion)	Major merger (equal-mass)
g_{grav}	7.86×10^{12}	2.95×10^{19} ($37.5\times$ larger)
$g_{\text{compressed}}$	1.0533×10^{22}	1.0533×10^{21} ($10\times$ larger)
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Title: NGC 4676 “The Mice”: UQFF Compressed Gravity Enhancement During Galaxy Merger – High-Velocity Tidal Bridge and $10\times$ Compression Factor

Author: Daniel T. Murphy

Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57)

Date: March 7, 2026

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Source Module: CondensedPhysics.py (NGC4676Model), validate_all_models.py
Index Slot: §1.7 arXiv Cross-Validation Framework, “PAPER_{0:D3}” -f[int]# PAPER #55 □ Mice Galaxies (NGC 4676): Merger Dynamics in UQFF
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Index Slot: §1.7 arXiv Cross-Validation Framework, PAPER_055

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UQFF Discovery: Novel application of UQFF calibration constants (? = $5.0 \times 10^{-4} \text{ day}^{-1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. System Parameters

Parameter	Value
Names	NGC 4676A + NGC 4676B (Arp 242)
Classification	The Mice (paired spirals, Coma constellation)
Distance	~ 87 Mpc ($z \sim 0.0220$)
Separation	~ 50 kpc (current projected)

Parameter	Value
Tidal tails	~160 kpc each (two symmetric tails)
Combined mass	~2×10 ¹¹ M _?
Stage	First pericenter passage (~160 Myr ago)
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2. The 10× Compression Enhancement

The NGC4676 UQFF model produces: - $g_{\text{compressed}} = 1.0533 \times 10^{?1}$ (10× larger than the standard $1.0533 \times 10^{?2}$) - $R_{\text{amplitude}} = 1.1586 \times 10^{?1}$ (10× larger than the standard $1.1586 \times 10^{?2}$)

This 10× enhancement is the UQFF signature of a **major merger event**. In the 26-layer compressed gravity framework:

$$g_{\text{compressed}}^{\text{merger}} = g_{\text{compressed}}^{\text{isolated}} \times \left(1 + \frac{\Delta M_{\text{overlap}}}{M_{\text{total}}} \right)^n$$

where $\Delta M_{\text{overlap}}$ is the mass in the overlapping region and $n \approx 2.3$ for head-on collisions. For The Mice, with ~30% mass overlap during pericenter: $(1 + 0.3)^{2.3} \approx 1.7$. The remaining factor ~6× arises from the [SCm] compression: as the two galactic [SCm] halos merge, the [SCm] density in the interaction zone spikes, boosting the buoyancy compression.

Combined: $1.7 \times \sim 6 \approx 10 \times$? consistent with the observed 10× $g_{\text{compressed}}$ enhancement.

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Test 1: Gravitational Field g_{grav}

The combined gravitational field of two merging galaxies at their mutual center of mass: - $g_{\text{grav}} = 2.9500 \times 10^{?10}$ m/s² (2× that of NGC3372 Carina, 37.5× that of UGC10214) - Physical basis: Two galaxies at 50 kpc separation and 87 Mpc distance produce a higher effective g_{grav} than any single system in the suite, except M42 (which is much more concentrated) - **PASS ?**

Test 2: Hubble Factor

- Hubble = 1.0002 ($z \approx 0.022$, modest cosmological correction)
- Matches local-universe result expected for ~87 Mpc distance
- **PASS ?**

Test 3: Compressed Gravity $g_{\text{compressed}}$

- $g_{\text{compressed}} = 1.0533 \times 10^{?1}$ ($10\times$ standard)
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- **PASS ?**

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- $R_{\text{amplitude}} = 1.1586 \times 10^{?1}$ ($10\times$ standard)
- Enhanced inter-galaxy MHD and acoustic resonance in the collisional plasma
- **PASS ?**

4. Physical Comparison: Mice vs. Tadpole

Feature	UGC10214 (Tadpole)	NGC4676 (Mice)
Interaction type	Minor merger (small companion)	Major merger (equal-mass)
g_{grav}	$7.86 \times 10^{?12}$	$2.95 \times 10^{?10}$ ($37.5\times$ larger)
$g_{\text{compressed}}$	$1.0533 \times 10^{?2}$	$1.0533 \times 10^{?1}$ ($10\times$ larger)
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Tail structure	One-sided 280 kpc tail	Two symmetric 160 kpc tails
UQFF dominance	Ug3 torque	[SCm] compression

The UQFF cleanly separates the two interaction types: - **Minor mergers** (Tadpole): Ug3 anisotropy ? one-sided tail, standard compression - **Major mergers** (Mice): [SCm] compression spike ? $10\times$ $g_{\text{compressed}}$, double symmetric tails

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Epoch	t (Myr ago)	UQFF state
Pre-pericenter	-200	$g_{\text{compressed}} = \text{standard}$, Ug4 active
Pericenter (now ~-160 Myr)	-160	$g_{\text{compressed}} \times 10$, [SCm] shock
Post-pericenter	0 (today)	$g_{\text{compressed}} \square 5\square 7\times$ (partially relaxed)
Final coalescence	+2500 Myr	$g_{\text{compressed}} ? \text{standard}$ (merged elliptical)

The UQFF model captures the snapshot at $t \approx -160$ Myr (post-pericenter relaxation) where the $10\times$ factor is appropriate.

Summary

Test	Quantity	Value	Status
1	g_{grav}	$2.9500 \times 10^{10} \text{ m/s}^2$	?
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UQFF computed: UQFF magnetic Jeans correction factor [SSq]×B²/(8p×?×c_s²) = 5.7e-1 × 1.3e-9 = 7.4e-10; Jeans mass deviation from standard = 7.4e-10 × M_J.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-8} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	10 Å^2	Inertia tensor scale
$E_{\text{react}}(0)$	10 eV	Reference reactive energy

A.2 F_U Master Equation (Complete 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()

